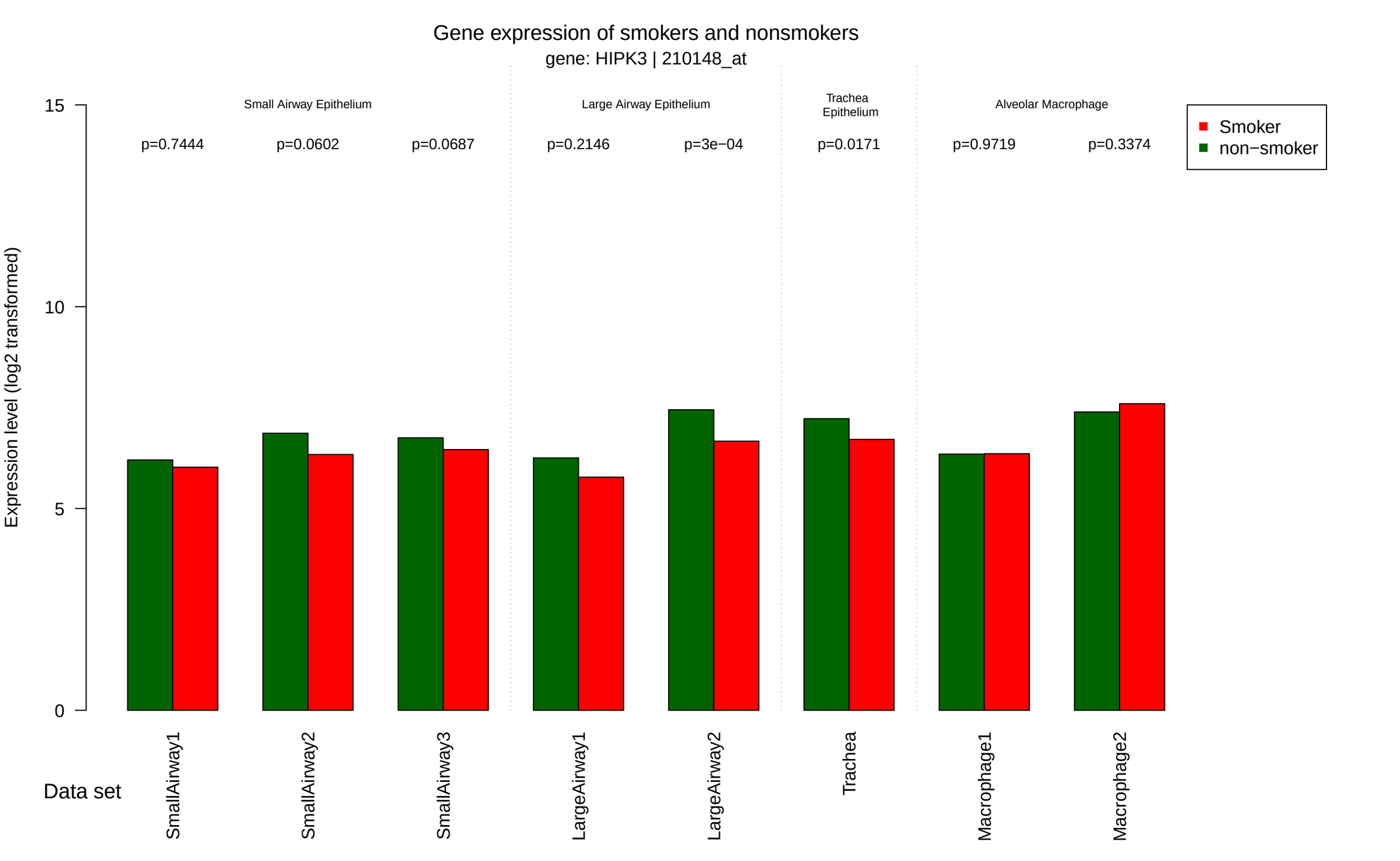
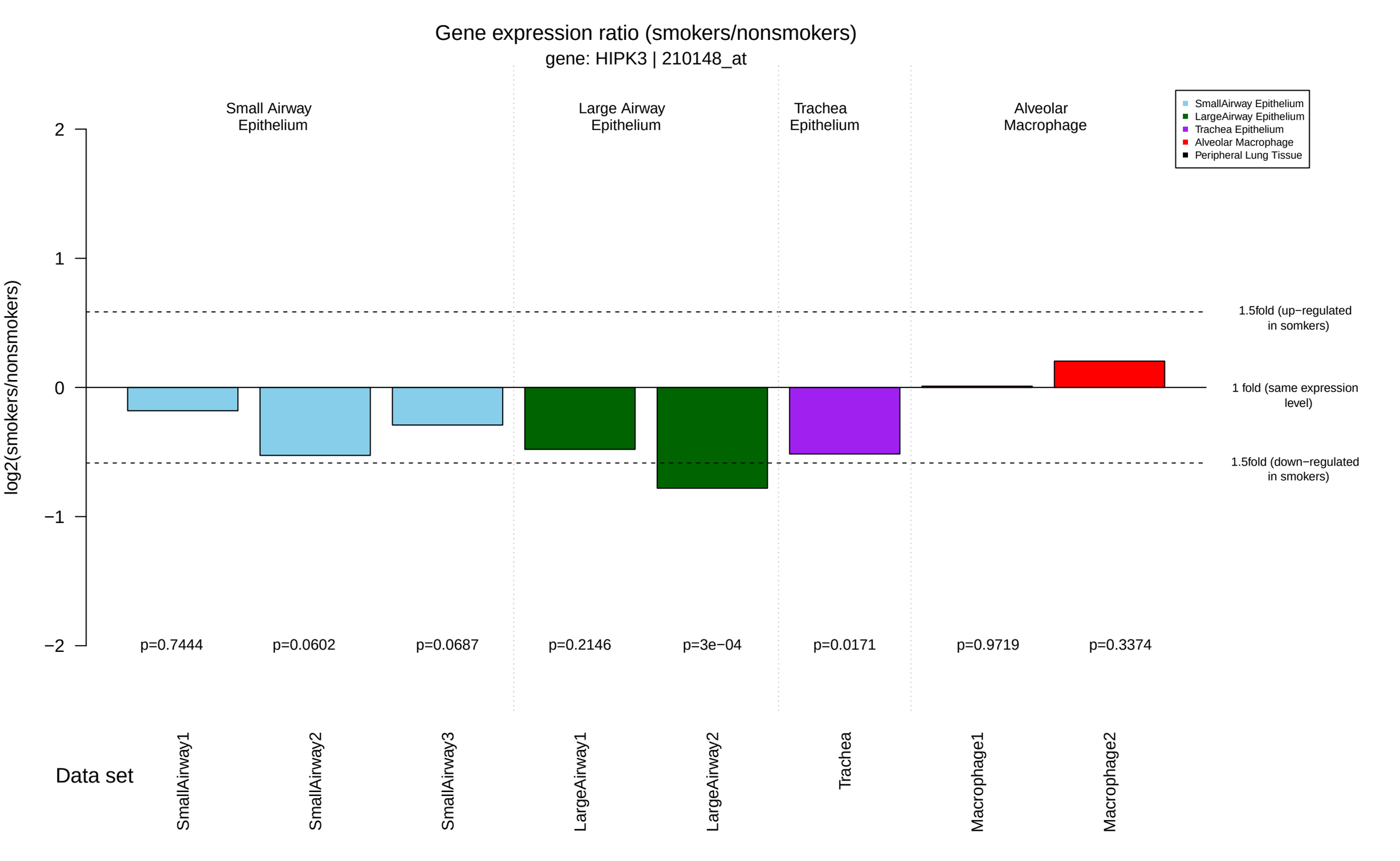
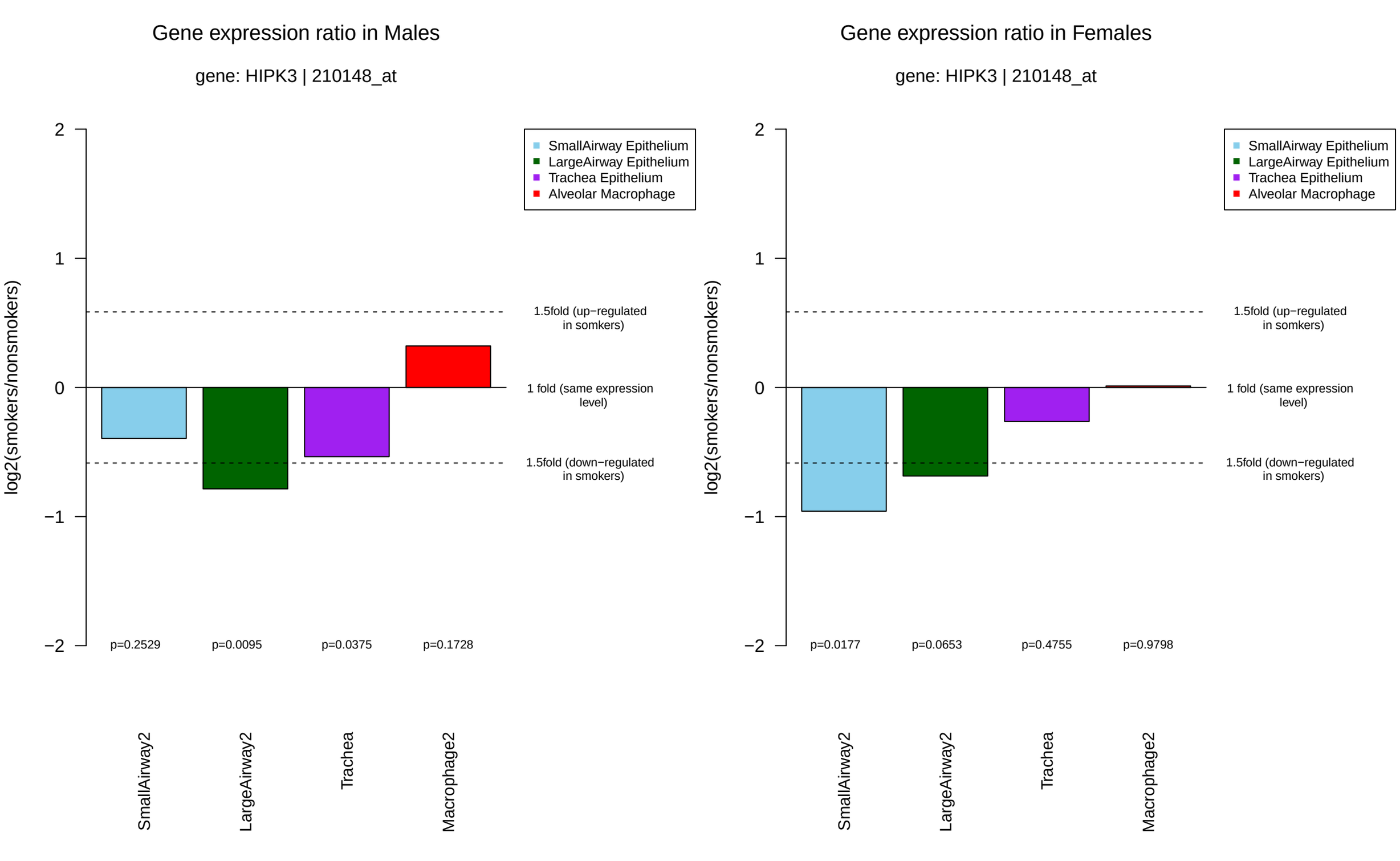
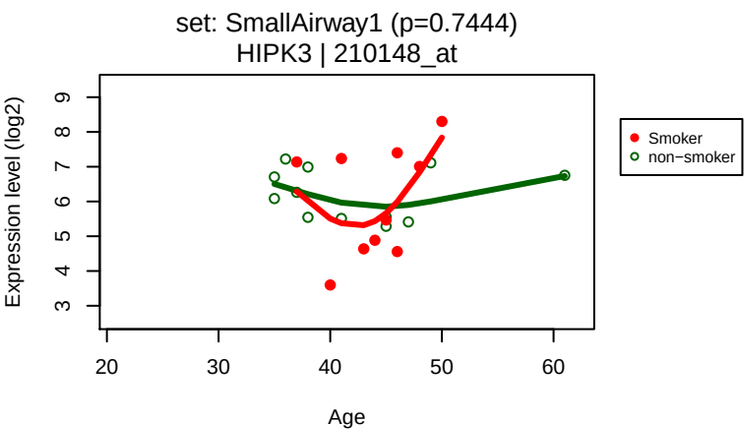
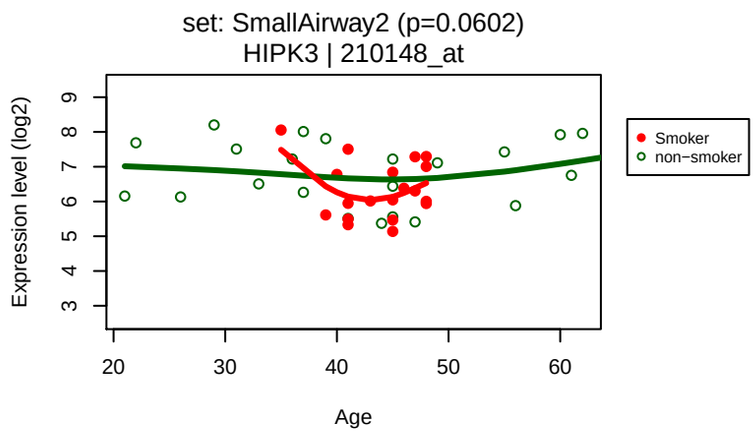
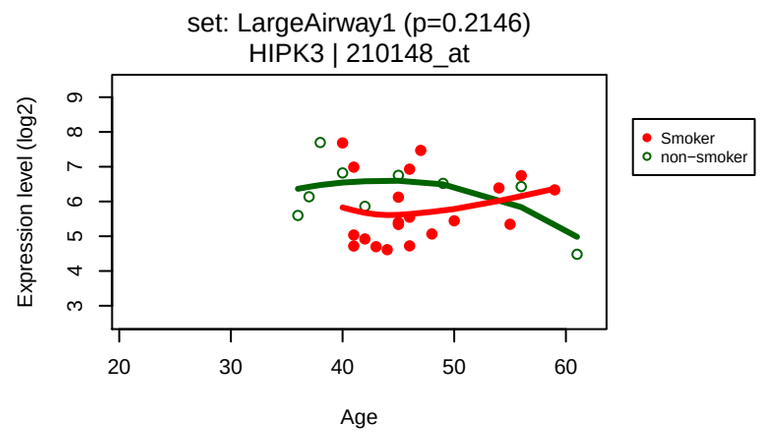

Gene expression of smokers and nonsmokers
gene: HIPK3 | 210148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7444
p=0.0602
p=0.0687
p=0.2146
p=3e−04
p=0.0171
p=0.9719
p=0.3374
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

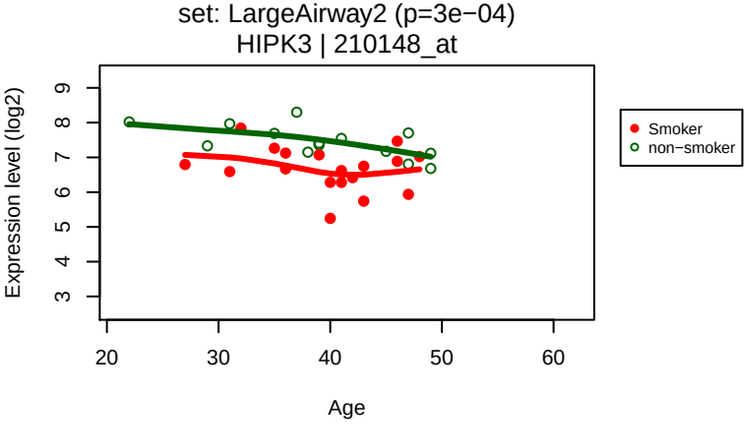

Gene expression ratio (smokers/nonsmokers)
gene: HIPK3 | 210148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7444
p=0.0602
p=0.0687
p=0.2146
p=3e−04
p=0.0171
p=0.9719
p=0.3374
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

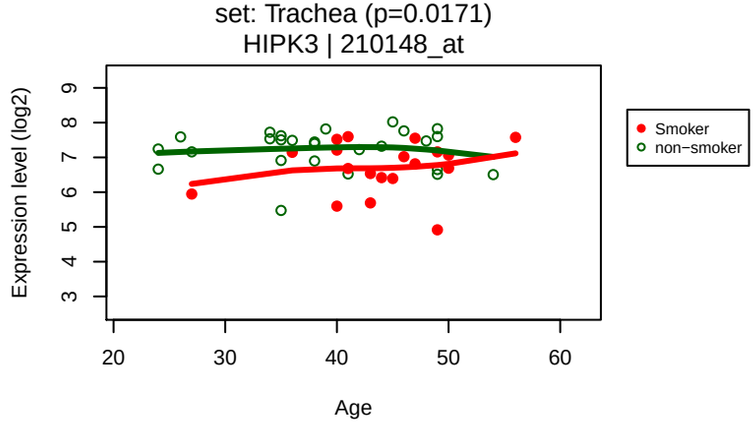

Gene expression ratio in Males
gene: HIPK3 | 210148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2529
p=0.0095
p=0.0375
p=0.1728
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK3 | 210148_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0177
p=0.0653
p=0.4755
p=0.9798
SmallAirway2
LargeAirway2
Trachea
Macrophage2

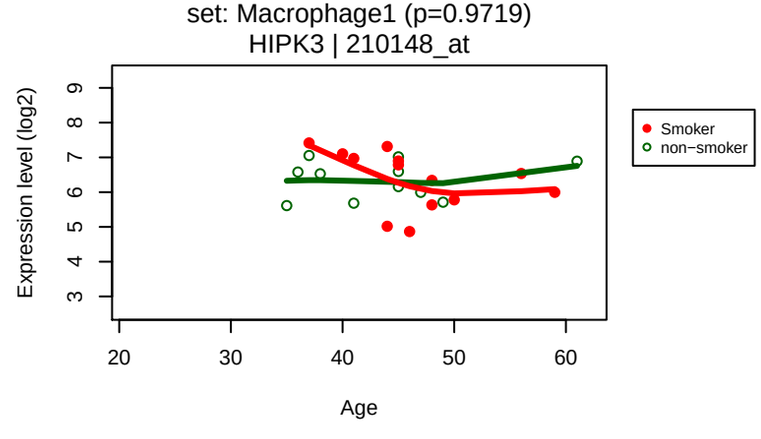

set: SmallAirway1 (p=0.7444)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker

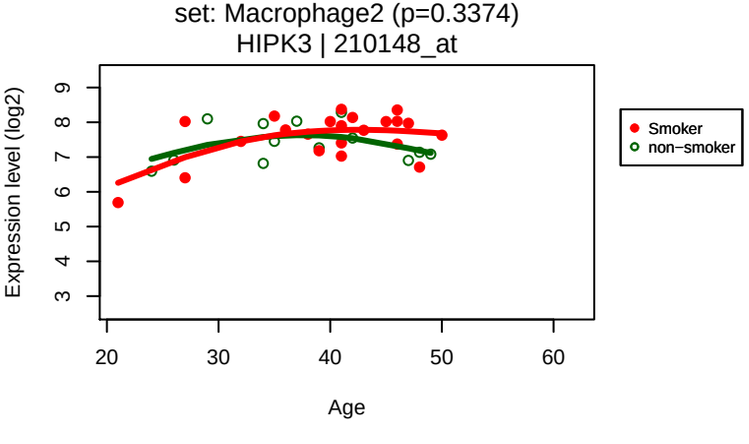

set: SmallAirway2 (p=0.0602)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker

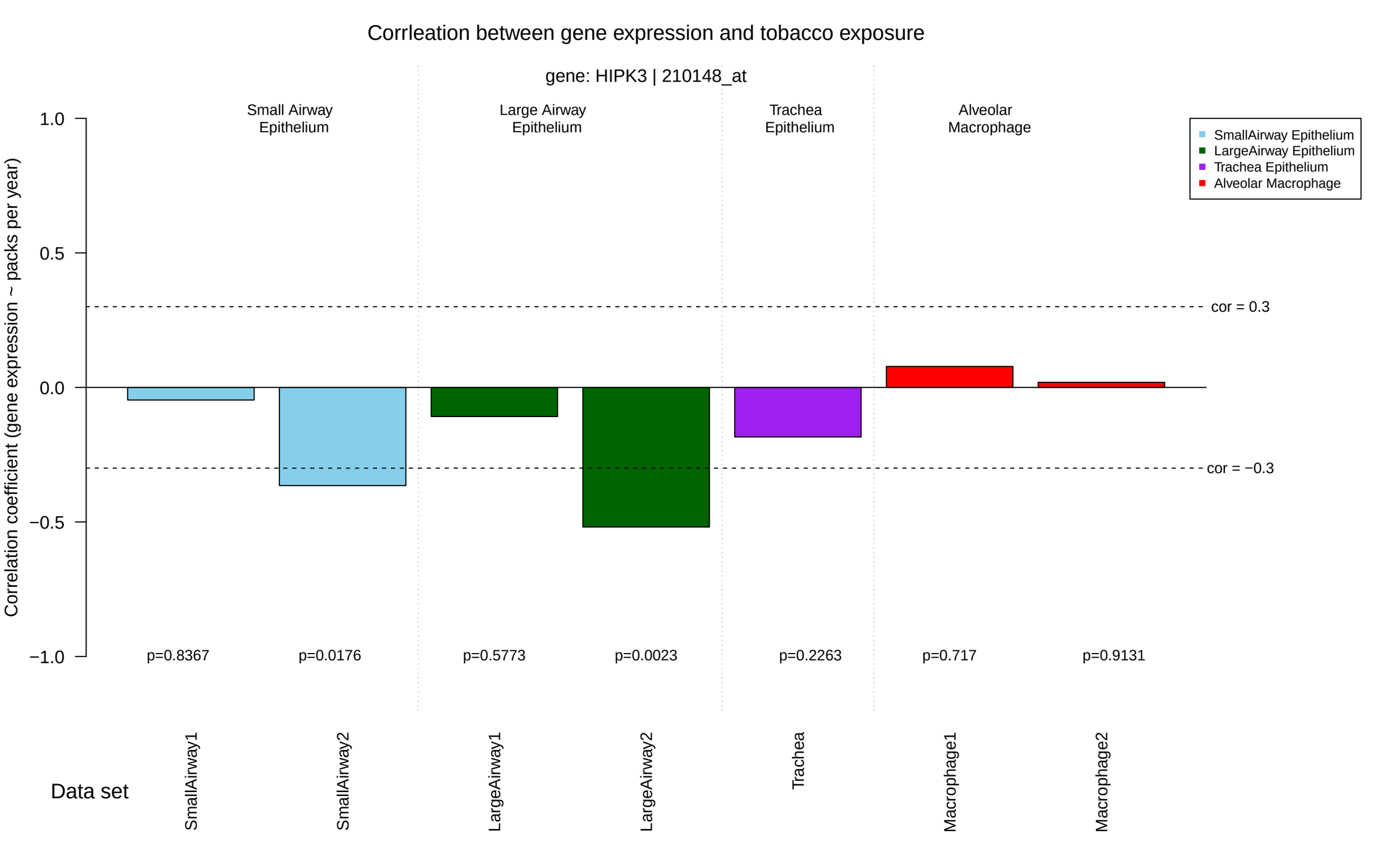

set: LargeAirway1 (p=0.2146)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=3e−04)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0171)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9719)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3374)
HIPK3 | 210148_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK3 | 210148_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8367
p=0.0176
p=0.5773
p=0.0023
p=0.2263
p=0.717
p=0.9131
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2